

Locally Robust Commensurate BART for Borrowing External Controls

Borrow where the sources agree, abstain where they do not, and report the map

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1. Established borrowing priors decide through one study-level parameter; whether an external patient is a fair control depends on who that patient is.
2. LRC-BART writes the RCT control surface as a pooled BART surface plus a sparse discrepancy ensemble, with a spike-and-slab commensurate prior on each discrepancy leaf.
3. The prior effective sample size is defined at the estimand and has a ceiling: the information the external source carries about the trial population.
4. Simulations with regional shifts, and a 30-patient single-arm myeloma trial with a registry control.

Potential gain

- More control observations
- Greater precision
- Smaller concurrent control arms

Why pooling can mislead

- Different eligibility and care pathways
- Different covariate distributions
- Measured and unmeasured confounding

External controls that match the trial well for one kind of patient can be badly off for another. A single borrowing parameter cannot tell them apart.

Where the established priors make the decision

Meta-analytic predictive prior (Neuenschwander et al., 2010; Schmidli et al., 2014)

Study parameters exchangeable,
 $\theta_j \mid \mu, \tau^2 \sim N(\mu, \tau^2)$; the robust version mixes in a vague component so that conflict switches borrowing off.

Commensurate prior (Hobbs et al., 2011, 2012)

$\theta \mid \theta_0 \sim N(\theta_0, 1/\tau)$ with a spike-and-slab hyperprior on the commensurability τ .

Power prior (Ibrahim and Chen, 2000)

Historical likelihood raised to a_0 .

Common feature

One heterogeneity or discount scale for the whole population, with no dependence on the patient's covariates. With a single historical study the three forms nearly coincide (Röver and Friede, 2025).

We make the borrowing decision region by region in covariate space, with the regions learned from the data.

The model in one line

RCT controls ($D_i = 1$) and RWD controls ($D_i = 2$), covariates $\mathbf{x}_i$, outcome y_i :

$$y_i = f(\mathbf{x}_i) + \mathbb{I}\{D_i = 1\} g(\mathbf{x}_i) + \varepsilon_i, \quad \varepsilon_i \sim N(0, \sigma_{D_i}^2)$$

f : external-control response mean

Standard BART, $H_f = 50$ trees, fitted to both sources. The RWD follow f .

g : discrepancy ensemble

Small and shallow, $H_g = 5$ trees, split prior $(0.5, 3)$, informed by the RCT controls only. The RCT controls follow $f_1 = f + g$.

- The decomposition is that of Bayesian causal forests (Hahn et al., 2020), with the source in the place of the treatment.
- The discrepancy is $g = f_1 - f$. Shrinking it toward zero centers the RCT-control mean f_1 at the external-control mean f .

The commensurate component is centered at $f(\mathbf{x})$

- $f(\mathbf{x})$ is the external-control response mean; $f_1(\mathbf{x})$ is the RCT-control response mean.
- Conditional on f , the discrepancy trees T_g and the scale, when every leaf reached by $\mathbf{x}$ is in the spike:

$$f_1(\mathbf{x}) \mid f, T_g, \{z_{h,\ell_h(\mathbf{x})} = 1\}_{h=1}^{H_g}, \tau_0^2 \sim \text{N} \left(\underbrace{f(\mathbf{x})}_{\text{external mean}}, H_g \tau_0^2 \right).$$

The same prior, expressed as a difference

Since $g(\mathbf{x}) = f_1(\mathbf{x}) - f(\mathbf{x})$, the corresponding discrepancy is centered at zero:

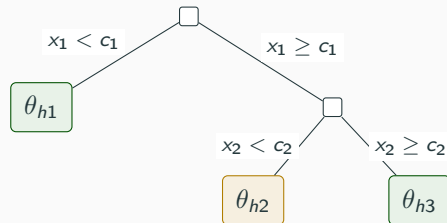
$$g(\mathbf{x}) \mid T_g, \{z_{h,\ell_h(\mathbf{x})} = 1\}_{h=1}^{H_g}, \tau_0^2 \sim \text{N}(0, H_g \tau_0^2).$$

H_g independent leaf contributions add their variances. These are priors for the latent response mean; residual variance is separate.

A shared surface with local discrepancies

RWD controls: $f(x)$ **RCT controls:** $f(x) + g(x)$

One tree in g (illustrative splits)



Each profile reaches one leaf per tree.
Add those contributions: $g(x) = \sum_{h=1}^{H_g} \theta_{h, \ell_h(x)}$.

f : the external response mean

Fitted jointly to both control sources.

Spike: $z_{h\ell} = 1$

$\theta_{h\ell} \sim N(0, \tau_0^2)$

Draws the RCT mean toward $f(x)$.

Slab: $z_{h\ell} = 0$

$\theta_{h\ell} \sim N(0, \tau_1^2)$, $\tau_1^2 > \tau_0^2$

Allows departures from $f(x)$.

$z_{h\ell} \sim \text{Bernoulli}(w)$. Leaf indicators vary;
 w, τ_0^2, τ_1^2 are shared.

The leaf prior of the discrepancy ensemble

The zero-centered mixture is on a leaf contribution to $g = f_1 - f$:

Each leaf (h, ℓ) carries

$$\theta_{h\ell} \mid z_{h\ell} \sim z_{h\ell} \text{N}(0, \tau_0^2) + (1 - z_{h\ell}) \text{N}(0, \tau_1^2), \quad z_{h\ell} \sim \text{Bernoulli}(w), \quad w \sim \text{Beta}(1, 1),$$

with $\tau_0^2 \sim \text{scaled-Inv-}\chi^2(\nu_0, s_0^2)$ truncated to $(0, \tau_1^2)$ and τ_1^2 fixed by the range rule.

Spike, $z = 1$: the commensurate component

Shrinks a discrepancy contribution toward zero, drawing the RCT response mean toward $f(x)$. The ESS calibration sets the leaf scale τ_0 .

Slab, $z = 0$: the vague component

A free local offset on the outcome scale: the robust-MAP vague component placed inside a leaf.

Special cases

$w = 1$: commensurate BART (C-BART), one learned scale. $w = 1, \tau_0^2 \rightarrow 0$: BART on the pooled controls (BART-CP). $w = 0$: a free source offset, the tree analogue of BART with the source indicator (BART-PP).

Why a separate discrepancy ensemble

The alternative looks equally natural: one ensemble of $H = 50$ trees with two means per leaf coupled by a common scale. Suppose the RWD surface is shifted by δ on a region.

- Shared leaves: the cheapest fit spreads δ/H across every tree under the spike at prior cost $\delta^2/(2H\tau^2)$; routing the shift through slab leaves wins only if $\delta > 2H\tau^2\sqrt{2c}/\lambda$, which at $H = 50$ is a shift of 17 to 86 on an outcome with range 10.
- Decomposition: the shift lands in one or two leaves of one g tree after a single split on the boundary, and the slab wins as soon as $\frac{\delta^2}{2}\{(H_g\tau_0^2)^{-1} - \tau_1^{-2}\}$ exceeds $\log \frac{w}{1-w} + \log \frac{\tau_1}{\tau_0}$.

Consequence

With fifty trees the spike can hide any clinically sized shift by spreading it, and the borrowing map stays flat. With few trees and a small τ_0 it cannot. H_g is kept small on purpose.

Every leaf marginal is exact

Conditionally on (τ_0^2, τ_1^2, w) the leaf prior is a finite Gaussian mixture, so each tree move uses closed-form factors:

$$M_f = (1 + \lambda_f^2 A)^{-1/2} \exp\left\{\frac{\lambda_f^2 B^2}{2(1 + \lambda_f^2 A)}\right\}, \quad M_g = \frac{w \phi(\bar{y}_1; 0, \tau_0^2 + v_1) + (1 - w) \phi(\bar{y}_1; 0, \tau_1^2 + v_1)}{\phi(\bar{y}_1; 0, v_1)},$$

with A, B the precision-weighted counts and sums of both sources in an f leaf, and $\bar{y}_1, v_1 = \sigma_1^2/n_1$ the RCT residual mean and its variance in a g leaf.

- Grow, prune and change moves use the exact collapsed ratios, so the sampler targets the exact posterior: no surrogate likelihood.
- Leaf draws are conjugate; z is drawn first, then θ ; τ_0^2 and w update from the spike leaves.
- Right-censored outcomes: log-normal AFT with data augmentation; nothing else changes.
- One sweep costs a standard BART on $n_1 + n_2$ observations plus a five-tree ensemble on n_1 .

Prior effective sample size at the estimand

Estimand: the RCT-standardized control mean $\mu = N^{-1} \sum_{i \leq N} f_1(\mathbf{x}_i)$. Conditionally on the RWD, $f_1 = f + g$ with $f \mid \text{RWD}$ the posterior of a BART fitted to the RWD alone and g from its prior. With the expected local-information-ratio ESS (Neuenschwander et al., 2020),

$$\text{ESS}_0(s_0^2) = \sigma_1^2 \mathbb{E}_{\tau_0^2} \left[\frac{1}{V_\mu^f + c_g H_g \tau_0^2} \right], \quad V_\mu^f = \text{Var}(N^{-1} \sum_i f(\mathbf{x}_i) \mid \text{RWD}).$$

The ceiling

As $s_0^2 \rightarrow 0$, $\text{ESS}_0 \rightarrow \sigma_1^2 / V_\mu^f$: the information the RWD posterior carries about μ . With matched covariate distributions it equals $n_2 \sigma_1^2 / \sigma_2^2$; covariate shift lowers it (Theorem 3(d)).

The Pr-rule

Stage 1 fits BART to the RWD once; Stage 2 selects the most informative spike scale whose ESS_0 respects N_{target} in 95% of Stage-1 blocks. A target may be absolute or a fraction of the ceiling.

A pointwise version $\text{ESS}(\mathbf{x}) = \sigma_1^2 / \{V_f(\mathbf{x}) + H_g \tau_0^2\}$ gives the local ESS map.

Three results

Leaf-level results condition on f , the g partitions and $(\sigma_1^2, \tau_0^2, \tau_1^2, w)$; $\bar{y}$ is the RCT residual mean of a leaf with n controls, $v = \sigma_1^2/n$.

Theorem 1: bounded local influence

The RWD-induced shift of a leaf estimate relative to the slab estimate satisfies $\sup_{\bar{y}} |I(\bar{y})| \leq \sqrt{2v \max\{\log C, \frac{1}{2}\}}$ and vanishes at a Gaussian rate as $|\bar{y}| \rightarrow \infty$. Under C-BART and BART-CP the shift is linear and unbounded.

Theorem 2: detection and the map

The spike probability is one half at $d_{1/2}$, a few multiples of $(\tau_0^2 + \sigma_1^2/n)^{1/2}$; the map $\mathcal{B}_c(x) = P(|g(x)| < c \mid \text{data}) \rightarrow \mathbb{I}\{|d| < c\}$ as $n \rightarrow \infty$, whereas the all-spike indicator does not settle.

Theorem 3: monotone, well-posed calibration

ESS_0 is continuous, convex and strictly decreasing in s_0^2 with limits σ_1^2/V_μ^f and 0; the Pr-rule has a unique solution, positive exactly when the target is below the ceiling, and the Monte Carlo rule recovers it.

The borrowing map

For each patient profile x the analysis reports

$$\mathcal{B}_c(x) = P(|g(x)| < c \mid \text{data}),$$

the posterior probability that the local discrepancy is below a threshold c fixed in advance on the outcome scale, together with the posterior mean and interval of $g(x)$ and the pointwise ESS.

- $c = 0.5$ in the simulations, one third of the RCT residual SD for Gaussian outcomes.
- The all-spike probability $B(x)$ is kept as a diagnostic: its prior scale is w^{H_g} and it does not settle as the RCT grows.
- In single-arm mode nothing informs g , the map is flat by construction, and the covariate-resolved display is the local ESS map.

What is proved and what is not

The map results are conditional on the g partitions and treat f as fixed; there is no statement about the posterior over partitions. Separation of shifted from compatible regions under the full posterior is shown empirically.

Simulation design

Common design

- $R = 500$ paired replicates
- $p = 10$ covariates, nonlinear surface
- 300 RWD controls
- Gaussian: RCT 200:100, $\sigma_1 = 1.5$, $\sigma_2 = 1.8$
- Survival: RCT 225:75, log-normal AFT, about a quarter censored
- Targets: absolute 50, 75, 100; fractional 0.25, 0.5, 0.9 of the ceiling

Five scenarios

- **Sc1** no confounding: source at random
- **Sc2** measured: source depends on (X_5, X_6)
- **Sc3** unmeasured: source and outcome depend on $\mathbf{U}$, observed as proxies at $\rho \in \{-0.5, 0, 0.5\}$
- **Sc4** regional shift (new): RWD surface raised by $\delta_2 \in \{0.5, 1, 2\}$ outside $\mathcal{R} = \{X_5 > 2\}$, half the population; secondary version on the null covariate X_7
- **Sc5** global shift (new): raised by δ_2 everywhere; the right answer is to abstain

Shifts chosen around the leaf resolution $d_{1/2} \approx 0.5$ of Theorem 2.

Comparators: LM/AFT and BART with no pooling, complete pooling or the source indicator; PSCL; MAP; hierarchical regressions; C-BART; LRC-BART ($w = 0$).

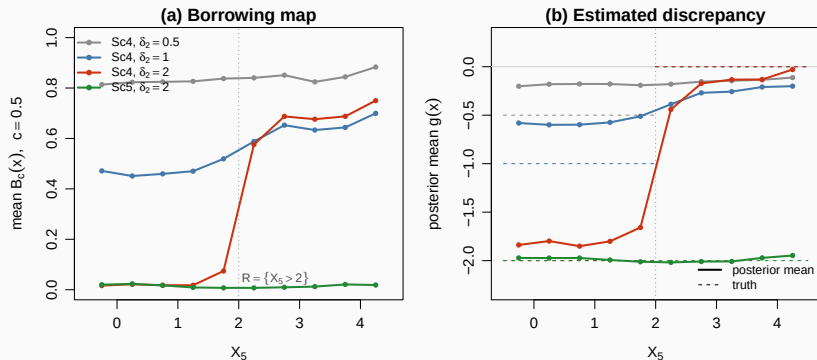
Gaussian outcomes: ATE bias, RMSE, coverage

Method	Sc1			Sc3, $\rho = 0$			Sc4, $\delta_2 = 1$			Sc4, $\delta_2 = 2$			Sc5, $\delta_2 = 2$		
	Bias	RMSE	Cov	Bias	RMSE	Cov	Bias	RMSE	Cov	Bias	RMSE	Cov	Bias	RMSE	Cov
LM-PP	-0.00	0.34	0.94	0.04	0.35	0.95	-0.01	0.34	0.95	-0.02	0.34	0.95	-0.04	0.34	0.94
BART-NP	0.00	0.22	0.91	-0.00	0.29	0.94	0.00	0.22	0.91	0.00	0.22	0.91	0.00	0.22	0.91
BART-CP	0.01	0.14	0.96	0.98	1.00	0.00	-0.34	0.37	0.35	-0.68	0.70	0.00	-1.39	1.40	0.00
BART-PP	0.01	0.21	0.94	0.03	0.27	0.94	-0.00	0.21	0.95	-0.01	0.21	0.96	-0.01	0.21	0.95
C-BART	0.01	0.15	0.95	0.23	0.38	0.87	-0.21	0.27	0.73	-0.23	0.39	0.66	-0.32	0.65	0.76
LRC-BART (100)	0.01	0.17	0.94	0.04	0.28	0.94	-0.04	0.22	0.89	0.00	0.20	0.92	-0.00	0.20	0.92

Reading the table

Agreement (Sc1): LRC-BART borrows, RMSE 0.172 against 0.205 for BART-PP; the pooling models win here. Global disagreement (Sc3, Sc5): it abstains, bias about 0.04 and 0.00, where C-BART and BART-CP fail. Regional shift: unbiased at $\delta_2 = 2$; at $\delta_2 = 1$ it borrows across the boundary, bias -0.04 , coverage 0.89.

The borrowing map under a regional shift



LRC-BART at target 100, first 40 replicates

At $\delta_2 = 2$ the map is 0.02 outside $\mathcal{R}$ and 0.7 inside, and g is -1.8 and -0.15 against -2 and 0 ; the transition occupies the unit around the boundary, learned from about four RCT controls per 0.1 of X_5 . At $\delta_2 = 1$ the estimate splits the shift ($-0.6 / -0.25$). Under Sc5 the map is 0.01 everywhere.

What the Gaussian study shows

When the sources agree

About 16% less RMSE than the free-offset model, and more on the control surface; realized ESS 53 against a prior ESS of 82 and a ceiling of 127.

When they disagree everywhere

Abstention: realized ESS 1.6 (Sc3) and 0.2 (Sc5), bias about 0.04 and 0.00; the 0.04 is 0.012 above BART-PP with paired SE 0.001, the price of a spike that is a Gaussian component and not a point mass.

When they disagree on a region

Found at 1.3 residual SDs with a one-unit boundary transition and a shortfall of a tenth of the shift on each side, a sample-size limit that tuning did not remove. At one residual SD, near $d_{1/2}$, the method over-borrows: bias -0.04 , coverage 0.89, RMSE 0.223 against 0.207; the control surface in $\mathcal{R}$ is not better than BART-PP's (0.824 against 0.785).

In Sc2 the ceiling falls from 127 to 23 under covariate shift, every absolute target is capped, and only the fractional ladder still moves.

Survival outcomes

Method	Sc1, median ratio			Sc3 $\rho = 0$, median ratio			Sc3 $\rho = 0$, RMST ratio		
	Bias	RMSE	Pow	Bias	RMSE	Cov	Bias	RMSE	Cov
AFT-NP	0.18	0.67	0.12	0.23	0.83	0.94	-0.03	0.15	0.92
BART-NP	0.26	0.51	0.46	0.19	0.57	0.90	0.06	0.17	0.92
BART-CP	0.06	0.23	0.63	2.31	2.41	0.00	0.59	0.61	0.01
BART-PP	0.03	0.35	0.25	0.04	0.45	0.91	-0.01	0.14	0.92
C-BART	0.03	0.25	0.42	0.58	0.91	0.88	0.14	0.22	0.85
LRC-BART (100)	0.05	0.31	0.34	0.15	0.56	0.94	0.03	0.15	0.94

Two readings

With 75 controls and censoring the median ratio has a wide, skewed posterior whose mean is biased upward for every method; borrowing makes it usable (Sc1: RMSE 0.312 against 0.345, power 0.34 against 0.25 for BART-PP). Under unmeasured confounding LRC-BART abstains (realized ESS 2.3) but its wider posterior, through the convex ratio, costs 0.11 of median-ratio bias against BART-PP; on the RMST ratio the biases are 0.03 and -0.005.

The single-arm design

Trial contributes treated subjects only ($n_T = 200$ Gaussian; 30 or 200 survival); the 300 RWD controls are the sole control source; LRC-BART in single-arm mode with $w = 1$.

No confounding

BART-CP and LRC-BART have Gaussian RMSE 0.16; the survival RMST ratio at $n_T = 30$ has RMSE 0.16 to 0.19 with coverage 0.92 to 0.97.

Measured confounding

Every method is biased: Gaussian ATE bias 1.68 (LM-CP), 0.41 (BART-CP), 0.42 (LRC-BART); nothing in the data can reveal a discrepancy where the RWD does not cover the trial profiles.

- The single-arm ceiling is 27 to 34 under Sc2, so an absolute target of 100 is capped and LRC-BART reduces to pooling; the fractional target still widens the interval (coverage 0.82 to 0.99).
- With $w < 1$ the slab only adds prior noise, variance $(1 - w)H_g\tau_1^2$: posterior SD 1.4 on the ATE and 10 to 34 on the median ratio with the point estimate unchanged. Hence $w = 1$ in single-arm mode.
- The hierarchical regressions with the stated hyperprior and no internal control are unusable (SD 8 on the ATE).

EloKRd phase II trial

- $n = 30$, all received EloKRd
- 13 PFS events; 10 deaths
- 50% high-risk cytogenetics; 30% ASCT

UCMM registry control

- $n = 253$ after cohort restrictions
- Standard induction regimens
- Six harmonized baseline covariates

A coding problem the calibration exposed

The non-Hispanic label differed between sources, so the earlier dummy coding predicted the 29 non-Hispanic trial patients from the 13-patient Hispanic registry stratum. The ESS ceiling at the trial profiles is 12.6 (PFS) and 14.6 (OS) under that coding and 63.7 and 36.7 under the harmonized one.

Estimand: standardized 5-year RMST ratio (neither trial median is reached). Kaplan–Meier: 1.015 for PFS, 0.90 for OS.

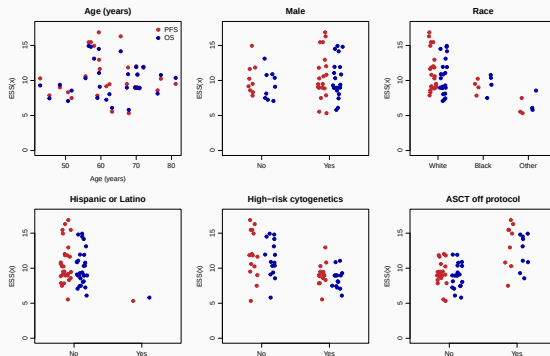
EloKRd results: 5-year RMST ratio, posterior mean and 95% interval

Harmonized coding	PFS	OS
Kaplan–Meier, unadjusted	1.01 (0.84, 1.22)	0.90 (0.79, 1.03)
Complete-pooling AFT	0.87 (0.69, 1.06)	0.80 (0.65, 0.94)
Hierarchical AFT	1.23 (0.74, 1.90)	1.21 (0.77, 1.70)
AFT-BART on UCMM, no discrepancy	0.97 (0.79, 1.16)	0.90 (0.78, 1.02)
LRC-BART, $N_{\text{target}} = 30$	0.97 (0.77, 1.23)	0.90 (0.77, 1.04)
LRC-BART, $N_{\text{target}} = 15$	0.99 (0.75, 1.40)	0.91 (0.76, 1.11)
LRC-BART, $0.9\times$ ceiling	0.97 (0.79, 1.15)	0.90 (0.77, 1.04)
<i>Earlier analysis, original coding: MAP-AFT-BART, 30</i>	1.08 (0.66, 16.4)	0.97 (0.74, 4.77)

Two readings

No nonparametric analysis supports a marginal EloKRd benefit ($P(\text{ratio} > 1)$ of 0.37 and 0.08); the interval widens as the target falls, on a scale of tenths. The long upper tails of the earlier analysis were a property of its prior: with the discrepancy carried by a five-tree ensemble of pointwise variance $H_g\tau_0^2 \approx 0.07$, the PFS ratio at target 30 has a 99.9th percentile of 1.6 over 8,000 draws.

The local ESS map at the 30 trial profiles



Reading the map

$ESS(x) = \sigma_1^2 / \{V_f(x) + H_g \tau_0^2\}$ ranges from 5 to 17 for PFS and 6 to 15 for OS: highest for White patients with ASCT and standard-risk cytogenetics, lowest for the one Hispanic patient, the Other/Unknown-race patients and the high-risk profiles without ASCT. A target of 30 is spread thinly over the enrolled profiles.

1. **Where borrowing is decided:** in the leaves of a discrepancy ensemble, so the tree prior chooses region by region and the choice is reported as a map.
2. **Exactness:** every leaf marginal is closed-form; the sampler targets the exact posterior.
3. **Calibration:** the prior ESS is defined at the estimand, has a ceiling equal to the information the external source carries about the trial population, and a target can be a fraction of that ceiling.
4. **Guarantees:** bounded local influence, leaf resolution and map consistency, monotone well-posed calibration.
5. **Evidence:** 16% RMSE gain under agreement, abstention under global conflict, detection of a regional shift of 1.3 SDs with a one-unit boundary transition, over-borrowing at a shift near the resolution.

Limitations and next steps

- Map results are conditional on the g partitions and on f fixed; nothing is proved about the posterior over partitions, which is where the boundary leak lives.
- A leaf-specific spike scale would let compatible regions differ, at the cost of conjugacy.
- Several external sources: one discrepancy ensemble per source with a common w ; the ceiling becomes a sum.
- Binary and count outcomes; nonparametric survival BART in place of the AFT model.
- Prospective calibration to Type I error rate and power, with the map as a pre-specified diagnostic.

Takeaway

LRC-BART borrows where the sources agree, abstains where they do not, and reports for each patient profile how much external information it used and how much was available.

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